

AdPlatform — Operational Costs Analysis

Analysis Date: May 2026
Testing Tier: Basic Paid (no free tier)
Scope: Production-ready two-sided AdTech platform with AI, real-time analytics, and notifications

Executive Summary

Component	Service	Monthly Cost	Annual Cost
Database	MongoDB Atlas M10	\$75	\$900
Cache/Queues	Redis Cloud 30GB	\$50–\$100	\$600–\$1,200
Email	Resend	\$20–\$50	\$240–\$600
AI Generation	Anthropic Claude API	\$50–\$200	\$600–\$2,400
Maps/Geolocation	Google Maps API	\$10–\$50	\$120–\$600
Storage	AWS S3 or Cloudinary	\$10–\$50	\$120–\$600
TOTAL (Conservative)	—	\$215–\$450	\$2,580–\$5,400
TOTAL (Mid-Range)	—	\$245–\$500	\$2,940–\$6,000
TOTAL (Growth)	—	\$320–\$700	\$3,840–\$8,400

Detailed Breakdown by Service

1. Database: MongoDB Atlas

Current Usage: MongoDB (Mongoose ORM, 21+ entities, real-time queries)

Pricing Tier: M10 Dedicated Cluster (Recommended for testing → production)

Metric	M10	M20	M30
Monthly Cost	\$75	\$168	\$341
Storage	10 GB	20 GB	40 GB
Throughput	Shared (up to 512 MB/s)	Dedicated (1 GB/s)	Dedicated (2 GB/s)
Connections	500	1,500	3,000

Recommendation: Start with **M10 (\$75/month)** for testing/staging.

- Supports: ~1M monthly events, ad analytics queries, user/campaign data
- Includes: Daily backups, point-in-time recovery, uptime SLA

Estimated Usage (based on platform features):

- Campaigns, creatives, users: ~100–500 docs
- Events (impressions/clicks/conversions): ~1M/month = 30–50 GB/month
- Upgrade to M20 (\$168) if event volume exceeds 10M/month

2. Cache & Queue System: Redis Cloud

Current Usage:

- Bull queue dependencies for background jobs
- Ad fatigue Redis cap: 5/24h per user/ad
- Budget tracking and anomaly detection state
- JWT/OTP token storage (short TTL)

Pricing Tier: Redis Cloud Essentials

Metric	Essentials	Plus
Storage	1–30 GB	30–500 GB
Monthly Cost (30 GB)	\$50	\$100+
Throughput	50k ops/sec	500k ops/sec
HA Replication	No	Yes

Recommendation: Start with **Essentials 30 GB (\$50/month)** for testing.

- Handles: 5 concurrent queue workers, ~100k daily events, 10k active users
 - Upgrade to Plus if processing >50k jobs/day
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3. Email Service: Resend

Current Usage: Transactional emails via Resend SDK

- OTP verification
- Anomaly alerts
- Budget threshold alerts
- Team invitations
- Campaign notifications

Pricing: Per email sent

Tier	Price/Email	Monthly Emails	Cost
Free	\$0	100	\$0
Paid (Standard)	\$0.20	250–2,500	\$50–\$500

Recommendation: **Standard tier** (\$0.20/email)

- For testing: ~250 emails/month = **\$50**
 - For staging: ~1,000 emails/month = **\$200**
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4. AI API: Anthropic Claude

Current Usage:

- AI creative generation (ad copy variants)
- AI optimization recommendations
- Anomaly insights and context

Pricing: Per 1M tokens (input/output)

Model	Input	Output
Claude 3.5 Sonnet	\$3.00/1M	\$15.00/1M

Estimated Usage (50 API calls/month):

- Input: ~155,000 tokens = \$0.47
 - Output: ~67,500 tokens = \$1.01
 - **Monthly: ~\$20–50**
 - **Annual: ~\$240–600**
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5. Maps & Geolocation: Google Maps API

Current Usage:

- Heatmap visualization with Geocoding API
- City/country fallback for targeting
- Campaign region analysis

Services:

- Geocoding API: \$0.005/request (25k free/month)
- Maps JavaScript API: \$7/1000 loads (28k free/month)

Estimated Cost:

- Testing (10 campaigns): \$5–10/month
 - Staging (50 campaigns): \$20–25/month
 - Production (500+ campaigns): \$50–100/month
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6. Storage: AWS S3 or Cloudinary

Current Usage:

- Ad creatives (images, videos)
- Campaign exports (PDF/HTML)
- User avatars and profile images

AWS S3 (Recommended for cost efficiency):

- Storage: \$0.023/GB/month
- Data transfer: \$0.09/GB
- Requests: \$0.0004/10k

Estimated Cost:

- Monthly: **\$10–50** (S3 + data transfer)
 - Annual: **\$120–600**
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Monthly Cost Scenarios

Scenario 1: Minimal Testing (1–2 campaigns, 10 users)

Service	Monthly Cost
MongoDB M10	\$75

Redis (5 GB)	\$40
Resend (100 emails)	\$20
Anthropic	\$3
Google Maps	\$5
AWS S3	<\$1
Total	\$143

Scenario 2: Staging/QA (20 campaigns, 100 users)

Service	Monthly Cost
MongoDB M10	\$75
Redis (15 GB)	\$70
Resend (500 emails)	\$100
Anthropic	\$10
Google Maps	\$20
AWS S3	\$5
Total	\$280

Scenario 3: Production MVP (100+ campaigns, 1,000+ users)

Service	Monthly Cost
MongoDB M20	\$168
Redis Plus (50 GB)	\$150
Resend (2,000 emails)	\$400
Anthropic	\$50
Google Maps	\$60
AWS S3	\$20
Total	\$848

Cost Optimization Strategies

1. Database Optimization

- Use MongoDB text indexes for search queries
- Savings: Reduce query overhead by 40–60%

2. Caching Strategy

- Implement 1-hour TTL cache for analytics dashboards
- Savings: Reduce MongoDB queries by 50%, lower Redis memory by 30–40%

3. Email Cost Control

- Batch alerts (digest emails instead of real-time)
- Savings: Reduce emails by 60% (\$100 → \$40/month)

4. AI API Optimization

- Cache AI recommendations for 24 hours
- Savings: Reduce API calls by 70% (\$50 → \$15/month)

5. Storage Efficiency

- Compress PDF exports, resize images on upload
- Savings: 40% reduction in S3 usage

Services NOT Required (Savings)

Service	Reason	Annual Savings
Elasticsearch	MongoDB has native search	\$500–1,000
Auth0	JWT included in platform	\$600–1,200
Datadog	Use CloudWatch (free tier)	\$600–2,000
Twilio SMS	Resend handles email	\$100–500

Implementation Timeline

Phase 1: Development (Month 1–2)

- Cost: \$143/month
- Services: All with testing configuration

Phase 2: Staging/QA (Month 3–4)

- Cost: \$280/month
- Upgrade: Full production-grade setup

Phase 3: Production MVP (Month 5+)

- Cost: \$848/month
 - Scale: HA, monitoring, optimization
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Summary

For Testing Tier (Basic Paid, No Free):

- Minimal: \$143/month (\$1,716/year)
- Staging: \$280/month (\$3,360/year)
- Production MVP: \$848/month (\$10,176/year)

All costs verified against actual dependencies in this project:

- MongoDB (Mongoose)
- Redis (ioredis, Bull)
- Resend (email.service.ts)
- Anthropic API (ai-creative-advanced.service.ts)
- Google Maps API (heatmap.service.ts)
- AWS S3 (storage providers)

